AMAN PARGANIHA

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Experience

Janyu Technologies — Vasai East, Maharashtra
Robotics Programming Intern — *Aug 2026 – Present*

- Building **MyFactory OS**, a manufacturing execution system for industrial-robotics production, combining an internal shop-floor job tracker with a customer-facing order-progress portal; owning backend delivery on a two-person team against a September MVP deadline.
- Authored the platform's technical scope after benchmarking 12+ MES and manufacturing-portal products (Siemens Opcenter, SAP ME, Tulip, Zetwerk); build-vs-buy case and MVP boundary approved by engineering leadership.
- Designed the relational schema and versioned REST API contract in **FastAPI, SQLAlchemy 2.0, PostgreSQL 16, Alembic** — JWT role-based access (admin/operator/customer), query-level tenant scoping, transactional stage-advance logic with QC gating, and an append-only audit log.
- Containerised the stack with **Docker Compose** (API, PostgreSQL, MinIO object storage, React/Vite frontend), keeping hosts and credentials environment-driven for on-premise deployment.

Education

International Institute of Information Technology Naya Raipur — India
M.Tech in Computer Science & Engineering (AI/ML) | CGPA: 8.0/10.0 — *Aug 2025 – Aug 2027*

Shri Shankaracharya Technical Campus Bhilai — India
B.Tech in Computer Science & Engineering | CGPA: 8.22/10.0 — *Aug 2019 – Aug 2023*

Technical Skills

Languages: Python, C++, SQL
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, vision-language models, reinforcement learning, NLP, fine-tuning, model evaluation & benchmarking
LLMs & Generative AI: RAG pipelines, LlamaIndex, LangChain, OpenAI API (GPT-4o / 4o-mini), prompt engineering, embeddings (text-embedding-3-large), vector databases (Qdrant), agentic workflows
Backend & MLOps: FastAPI, REST API design, SQLAlchemy, Alembic, PostgreSQL, JWT auth, Docker Compose, AWS (EC2), MinIO/S3, Inngest, Streamlit, Git/GitHub, CI
Data & Practices: Pandas, NumPy, ETL pipelines, dataset curation, feature engineering, SQL/NoSQL, Matplotlib, unit & end-to-end testing (Playwright), code review, system design

Projects

Event-Driven RAG Agent for Document Ingestion & Q&A | Code | *Python, FastAPI, Inngest, LlamaIndex, Qdrant, OpenAI, Docker*

- Built an event-driven RAG system (FastAPI backend, Streamlit chat UI, Dockerized Qdrant) that asynchronously ingests multi-page PDFs and answers natural-language queries with grounded, citation-backed responses.
- Architected durable, fault-tolerant ingestion with Inngest (independently retryable parse → embed → store steps) over LlamaIndex chunking, OpenAI `text-embedding-3-large` embeddings and a Qdrant vector store, kept modular across chunking/embedding strategies.

Multimodal Credit Risk Analysis using SEC XBRL & NLP | Code | *Python, Pandas, Scikit-learn, NLTK/VADER*

- Engineered an end-to-end pipeline parsing ∼41M raw SEC XBRL records (2022–2024, 12 quarters) into a curated dataset of ∼35,000 companies with 47 features (34 financial + 13 NLP-derived).
- Benchmarked Random Forest, Gradient Boosting, Logistic Regression and SVM for credit-rating prediction, reaching 97.9% binary / 93.8% multi-class accuracy on financial features; ran controlled per-feature-set evaluation and data-leakage checks to invalidate an inflated multimodal result.

Research

LiveJEPA: Real-Time Vision-Language Captioning (M.Tech thesis, in progress; PyTorch) — reconstructing Meta FAIR's VL-JEPA pipeline for streaming video captioning: frozen V-JEPA 2 encoder with a lightweight predictor head aligned to the SONAR text-embedding space; evaluating on EgoExo4D.
GreenCloudRL: Meta-RL for Energy-Efficient Cloud Scheduling (manuscript submitted) — hierarchical meta-RL scheduler (A2C/PPO with Reptile) with SHAP-based explainability, achieving ∼14% energy reduction on Google and Alibaba cluster traces.

Open Source Contributions

Matplotlib — PR #31422 merged into main and shipped in v3.11.0: resolved issue #26620 by improving legend parameter documentation; reviewed and approved by core maintainer *timhoffm*.
JupyterLab — PR #18610 merged, PR #18668 approved: improved user-facing documentation and added Playwright end-to-end test coverage; both cleared multi-round maintainer review.

Certifications & Achievements

GATE CSE: Top 7.8% of ∼210,000 candidates nationwide — secured M.Tech (AI/ML) at IIIT Naya Raipur. | **Problem Solving:** 200+ LeetCode problems solved.
Google × Kaggle — 5-Day AI Agents Intensive: Agentic LLM workflows, LlamaIndex/LangChain orchestration, RAG system design.